Project 2

Group Members: Binh & Oliver

Accessing data

For our analysis, we have chosen to go with 3 csv files containing answers to an online survey regarding subjective judgements of likeliness. The `absolute` dataset contains responses to a task asking respondents to rank various statements about probability, like **might happen**, **little chance**, etc… on a scale from 0 (not happening at all) to 100 (certainly would happen). Additionally, the `pairwise` dataset contains responses to "which term represents a higher probability" questions. Finally, the `respondents` dataset contains some simple demographic questions. The dataset is available as a part of `Tidytuesday`, and can be accessed using this link.

```
absolute <- readr::read_csv('https://raw.githubusercontent.com/rfordatascience/tidytuesday/ma
pairwise <- readr::read_csv('https://raw.githubusercontent.com/rfordatascience/tidytuesday/ma
respondents <- readr::read_csv('https://raw.githubusercontent.com/rfordatascience/tidytuesday
```

A good data scientist always checks the data they import.

```
summary(absolute)
```

```
  response_id       term             probability        order
 Min.   :   4   Length:98306       Min.   :  0.00   Min.   : 1
 1st Qu.:1297   Class :character   1st Qu.: 10.00   1st Qu.: 5
 Median :2590   Mode  :character   Median : 50.00   Median :10
 Mean   :2590                      Mean   : 45.56   Mean   :10
 3rd Qu.:3884                      3rd Qu.: 75.00   3rd Qu.:15
 Max.   :5177                      Max.   :100.00   Max.   :19
```

```r
summary(pairwise)
```

```
  response_id      pair_id          term1               term2          
 Min.   :   4   Min.   : 1.0   Length:51740       Length:51740      
 1st Qu.:1297   1st Qu.: 3.0   Class :character   Class :character  
 Median :2590   Median : 5.5   Mode  :character   Mode  :character  
 Mean   :2590   Mean   : 5.5                                        
 3rd Qu.:3884   3rd Qu.: 8.0                                        
 Max.   :5177   Max.   :10.0                                        
   selected        
 Length:51740      
 Class :character  
 Mode  :character  
```

```r
summary(respondents)
```

```
  response_id    timestamp           age_band         english_background
 Min.   :   4   Length:5174        Length:5174        Length:5174       
 1st Qu.:1297   Class :character   Class :character   Class :character  
 Median :2590   Mode  :character   Mode  :character   Mode  :character  
 Mean   :2590                                                           
 3rd Qu.:3884                                                           
 Max.   :5177                                                           
 education_level    country_of_residence
 Length:5174        Length:5174         
 Class :character   Class :character    
 Mode  :character   Mode  :character    
```

Since we accessed the data using links, we have to write the csvs out first.

```r
write_csv(absolute, "absolute.csv")
write_csv(pairwise, 'pairwise.csv')
write_csv(respondents, 'respondents.csv')
```

Perusal of the dataset shows that the `absolute` and `pairwise` datasets do not allow for NA entries at all. On the other hand, text fields for `respondents` do allow for missing data. This is important for database creation purposes, which we will use `duckdb` for.

Designing the database

Now we invoke `duckdb` and establish a connection to our new database.

```
con_prob <- DBI::dbConnect(duckdb::duckdb(), dbdir = "probability_db")
```

```
USE probability_db
```

```
DROP TABLE IF EXISTS absolute;
DROP TABLE IF EXISTS pairwise;
DROP TABLE IF EXISTS respondents;
```

We now create the tables on the database. Some motivation for the codechunk below:

- given that this is only 2 months worth of observations, going with INTEGER for response_id seems reasonable.
- other numeric variables only go from 0-100 and 1-19
- in `absolute`, each respondent will get presented with a term once –> unique row. Additionally, a valid row would need to have a respondent, and a term they are presented with to rank. This means that while `probability` can be NULL, other terms cannot.
- in `pairwise`, each participant gets presented with a pair of terms only once –> unique row. Similarly, a respondent might not choose a term between a presented pair, but each row would need to have a respondent identifier, and contain info about the pair they are presented with.
- in `respondents`, there are columns where 255 characters were not needed (like `timestamp` or `country_of_residence`). I've chosen to limit the character count to 60 for those columns.

```sql
CREATE TABLE absolute (
  response_id INTEGER NOT NULL,
  term VARCHAR(255) NOT NULL,
  probability TINYINT DEFAULT NULL,
  "order" TINYINT NOT NULL,
  PRIMARY KEY (response_id, term)
);

CREATE TABLE pairwise (
  response_id INTEGER NOT NULL,
  pair_id INTEGER NOT NULL,
  term1 VARCHAR(255) NOT NULL,
  term2 VARCHAR(255) NOT NULL,
  selected VARCHAR(255) DEFAULT NULL,
  PRIMARY KEY (response_id, pair_id)
);

CREATE TABLE respondents (
  response_id INTEGER NOT NULL,
  "timestamp" VARCHAR(60) NOT NULL,
  age_band VARCHAR(60) DEFAULT NULL,
  english_background VARCHAR(255) DEFAULT NULL,
  education_level VARCHAR(60) DEFAULT NULL,
  country_of_residence VARCHAR(60) DEFAULT NULL,
  PRIMARY KEY (response_id)
);
```

Having created the tables, we can now copy the contents of the csv's onto the tables in our database:

```sql
COPY absolute FROM 'absolute.csv' (FORMAT CSV, HEADER, NULL 'NA');
COPY pairwise FROM 'pairwise.csv' (FORMAT CSV, HEADER, NULL 'NA');
COPY respondents FROM 'respondents.csv' (FORMAT CSV, HEADER, NULL 'NA');
```

The fingers are crossed, and the database seems to be working properly.

Querying data using SQL

Binh's queries

Query 1: examining the distribution of education background among respondents

```sql
SELECT education_level, COUNT(*) AS n
FROM respondents
WHERE education_level IS NOT NULL
GROUP BY education_level
ORDER BY n DESC
```

Table 1: 5 records

education_level	n
Postgraduate	2699
Bachelor	1437
Some college	263
High school	103
Less than high school	9

The distribution seems to be a little lopsided - respondents from a high educational background seems to be overrepresented in this dataset compared to the population average.

Query 2: examining the highest (absolute) probability assigned to each term

```sql
SELECT term, MAX(probability) AS max_prb
FROM absolute
GROUP BY term
```

Table 2: Displaying records 1 - 10

term	max_prb
Likely	100
Realistic Possibility	100
Remote Chance	100
Very Good Chance	100

term	max_prb
About Even	100
Better than Even	100
May Happen	100
Highly Unlikely	100
Unlikely	100
Almost Certain	100

Given that this is a small subset of the online survey, it's reasonable to assume that there might be some data quality concerns.

Query 3: which terms win the most often in pairwise comparisons?

```
SELECT selected AS term, COUNT(*) AS n_wins
FROM pairwise
GROUP BY selected
ORDER BY n_wins DESC
```

Table 3: Displaying records 1 - 10

term	n_wins
Will Happen	5274
Highly Likely	4771
Almost Certain	4700
Very Good Chance	4458
Probable	4128
Likely	3911
Better than Even	3633
Realistic Possibility	3485
About Even	2689
May Happen	2641

Pairwise comparisons jointly create a reasonable order of terms(?)

Query 4: which terms have the highest variance in absolute probability among American respondents?

```
SELECT term, VAR_SAMP(probability) AS var_prob
FROM absolute JOIN respondents ON absolute.response_id = respondents.response_id
```

```sql
WHERE country_of_residence = 'United States'
GROUP BY term
ORDER BY var_prob desc
```

Table 4: Displaying records 1 - 10

term	var_prob
Realistic Possibility	413.6271
Could Happen	316.8297
Might Happen	284.6066
May Happen	268.9883
Highly Unlikely	170.7584
Improbable	158.4131
Probable	148.2369
Unlikely	139.4656
Chances are Slight	125.6742
Highly Likely	112.6015

Very illuminating: before planning something, it might be good to ensure you and your friend align on what a "realistic probability" actually is.

Oliver's Queries:

Query 1: How consistent is each respondent in their probability ratings?

```sql
SELECT response_id, AVG(probability) AS AvgRating, STDDEV(probability) AS ProbStdDev, COUNT(
FROM absolute
WHERE probability IS NOT NULL
GROUP BY response_id
ORDER BY ProbStdDev;
```

Table 5: Displaying records 1 - 10

response_id	AvgRating	ProbStdDev	n
4572	0.000000	0.0000000	19
1701	1.052632	0.4046513	19
139	80.526316	6.4322833	19
2512	64.210526	11.9423567	19
4672	46.052632	12.8645667	19

response_id	AvgRating	ProbStdDev	n
3855	77.894737	14.1731151	19
5007	82.526316	17.0044714	19
5164	66.578947	17.2572554	19
4331	76.052632	17.8153295	19
2341	50.263158	19.2031308	19

Interesting - it seems that some respondents are very consistent in their responses, like 1701 and 139, yet the range of consistency is vast. Confirms that responses are somewhat random and arbitrary.

Query 2: How many times did each term show up in a pairwise comparison?

```
SELECT term1 AS term, COUNT(*) AS n
FROM pairwise
GROUP BY term
ORDER BY n DESC;
```

Table 6: Displaying records 1 - 10

term	n
Remote Chance	3024
About Even	3020
Improbable	2914
Almost No Chance	2894
Highly Unlikely	2867
Better than Even	2771
Realistic Possibility	2764
Likely	2740
Little Chance	2735
Might Happen	2725

This tell us that Binh's Query 3 result is further valid as the distribution is mostly fair across pairwise comparisons. It also tells us that the survey was pretty fair in the number of times each term showed up as term 1, which does impact bias.

Query 3: How old are the respondents? What age group responded most?

```
SELECT age_band AS age_group, COUNT(*) AS n
FROM respondents
WHERE age_group IS NOT NULL
GROUP BY age_group
ORDER BY n DESC;
```

Table 7: 8 records

age_group	n
35-44	1405
45-54	1163
25-34	868
55-64	734
65-74	328
18-24	141
75+	79
Under 18	19

Mainly middle aged respondents. Makes sense as younger people are probably less likely to fill out a survey. That being said, it means the findings from the survey are going to be reflective of middle-aged adults' perception on the words, not necessarily the whole population (35-54yr olds make up roughly 54% of respondents).

Query 4: Do different ages have different average probability ratings?

```
SELECT r.age_band AS age_group, AVG(a.probability) AS AvgProb
FROM absolute a
JOIN respondents r ON a.response_id = r.response_id
WHERE r.age_band IS NOT NULL AND a.probability IS NOT NULL
GROUP BY age_group
ORDER BY AvgProb DESC;
```

Table 8: 8 records

age_group	AvgProb
Under 18	48.67313
18-24	46.17245
35-44	45.92943
25-34	45.88328
75+	45.74950

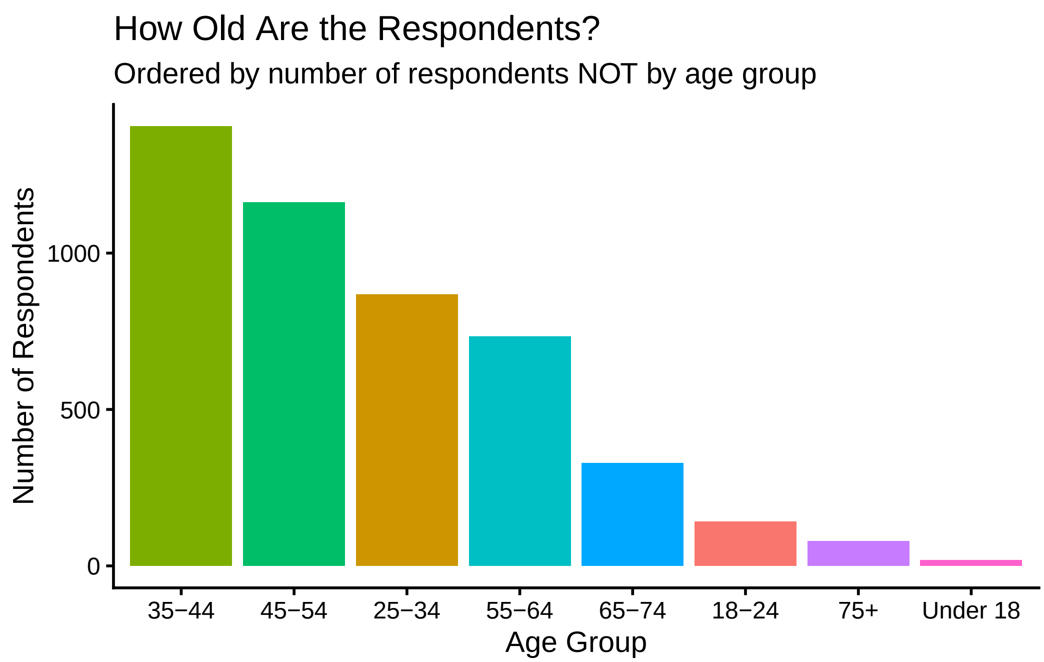
age_group	AvgProb
45-54	45.25967
65-74	45.19320
55-64	44.69188

note: have to use original name of variable or an error message is thrown.

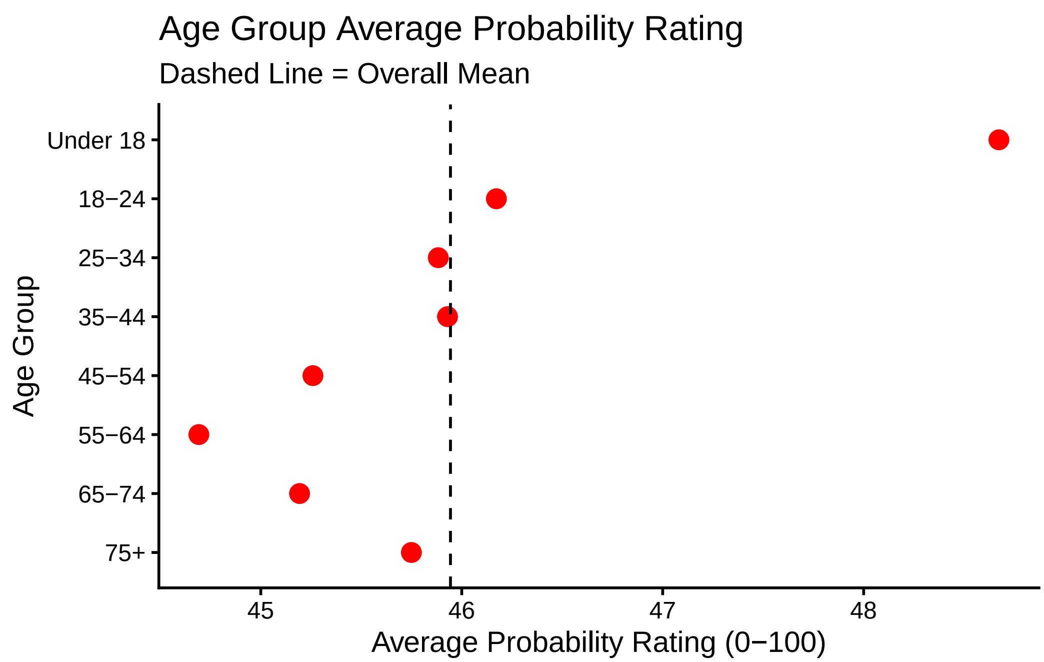
analysis: In contrast to my last query, it seems age has little effect on how the terms are perceived. As such, it isn't AS much of a worry that the age of respondents is highly concentrated on middle aged adults.

Creating data visualizations

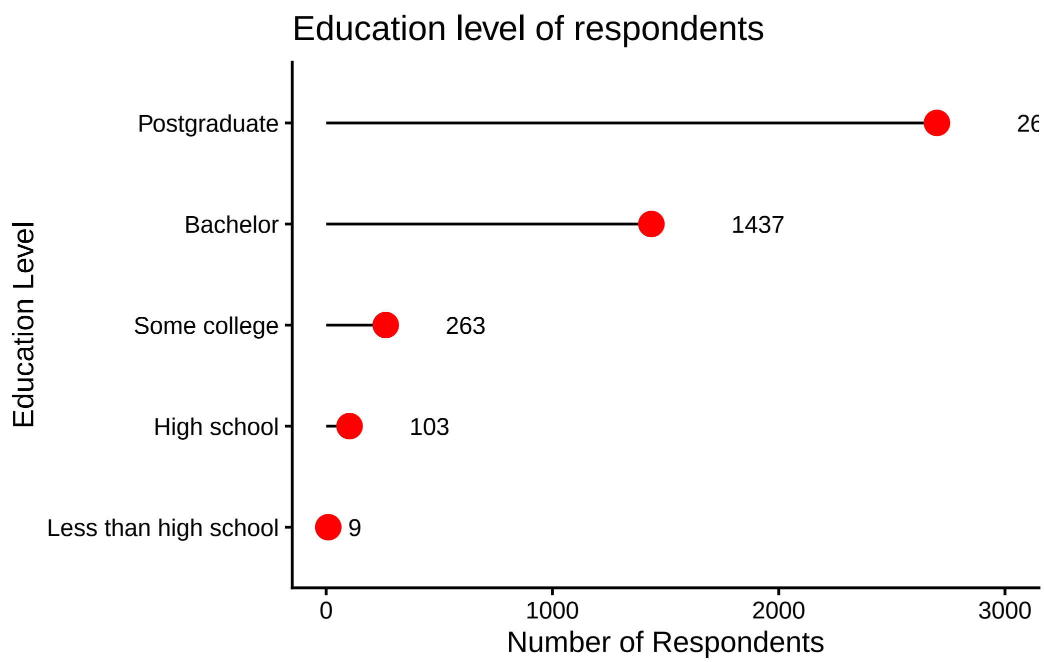
```
vis1 <- dbGetQuery(con_prob,
"SELECT age_band AS age_group, COUNT(*) AS n
FROM respondents
WHERE age_band IS NOT NULL
GROUP BY age_band
ORDER BY n DESC")

ggplot(vis1, aes(x = reorder(age_group, -n), y = n, fill = age_group)) +
  geom_col(show.legend = FALSE) +
  labs(
    title = "How Old Are the Respondents?",
    subtitle = "Ordered by number of respondents NOT by age group",
    x = "Age Group",
    y = "Number of Respondents",
  ) +
  theme_classic()
```




```
vis2 <- dbGetQuery(con_prob,
"SELECT r.age_band AS age_group, AVG(a.probability) AS AvgProb
FROM absolute a
JOIN respondents r ON a.response_id = r.response_id
WHERE r.age_band IS NOT NULL AND a.probability IS NOT NULL
GROUP BY age_group
ORDER BY AvgProb DESC")

vis2 <- vis2 |>
  mutate(age_group = factor(age_group, levels = c("75+", "65-74", "55-64", "45-54", "35-44",

ggplot(vis2, aes(x = AvgProb, y = age_group)) +
  geom_point(size = 3, color = "red") +
  geom_vline(xintercept = mean(vis2$AvgProb), linetype = "dashed", color = "black") +
  labs(
    title = "Age Group Average Probability Rating",
    subtitle = "Dashed Line = Overall Mean",
    x = "Average Probability Rating (0-100)",
    y="Age Group"
  ) +
  theme_classic()
```




```
vis3 <- dbGetQuery(con_prob,
"SELECT education_level, COUNT(*) AS n
FROM respondents
WHERE education_level IS NOT NULL
GROUP BY education_level
ORDER BY n DESC")

ggplot(vis3, aes(x = n, y = reorder(education_level, n))) +
  geom_segment(aes(x = 0, xend = n,
                   y = reorder(education_level, n),
                   yend = reorder(education_level, n)), color = "black") +
  geom_point(size = 4, color = "red") +
  geom_text(aes(label = n), hjust = -1.5, size=3) +
  scale_x_continuous(limits=c(0,3000)) +
  labs(
    title = "Education level of respondents",
    x="Number of Respondents",
    y="Education Level",
  ) +
  theme_classic()
```



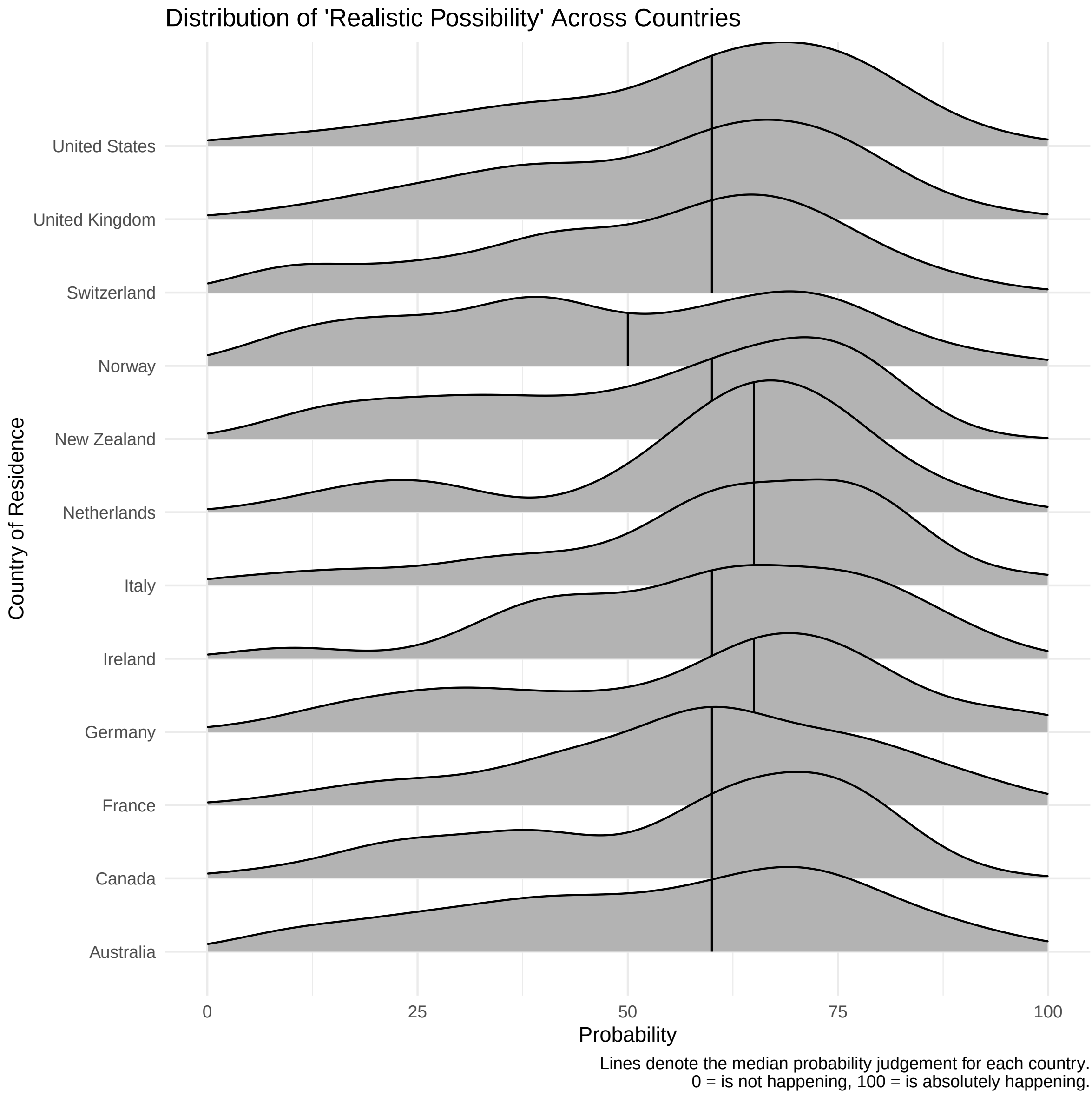
Binh's visualization: How variable is "Realistic Probability", actually?

- First, we pull the query from `dbGetQuery`.
- Then, we use Ridgeline Plots from `ggridges()` to get a better feel for the distribution of "Realistic Possibility" across different countries. Boxplots were initially considered, but ultimately discarded in favor of Ridgeline plots, since the latter is better able to visualize the actual distribution of probabiilty judgements.

```
bData <- dbGetQuery(con_prob, "SELECT term, respondents.response_id, probability, country_of_

# Initial graph didn't look too hot, so filtering to countries w more than 30 responses
bData30 <- bData |>
  group_by(country_of_residence) |>
  filter(n() > 30) |>
  ungroup()

ggplot(bData30, aes(x = probability, y = country_of_residence)) +
  labs(x = "Probability",
       y = "Country of Residence",
```

```r
      title = "Distribution of 'Realistic Possibility' Across Countries",
      caption = "Lines denote the median probability judgement for each country.\n 0 = is no
  xlim(0,100) +
  geom_density_ridges(quantile_lines = TRUE, quantiles = 2) +
  theme_minimal()
```

```
Picking joint bandwidth of 7.15
```



We can see that the distribution of probability judgments for the term "Realistic Possibility" is incredibly wide across all countries, ranging between all possible values on the scale. Interestingly, there seems to be some convergence on the median judgment value of the term - most countries seem to have a median probability of about 60 for the term.

Disconnect

```r
DBI::dbDisconnect(con_prob, shutdown = TRUE)
```